

|                           |                                       |                        |                       |
|---------------------------|---------------------------------------|------------------------|-----------------------|
| <b>COURSE CODE/TITLE:</b> | CPE212 - Automating Server Management | <b>SECTION:</b>        | CPE31S2               |
| <b>DATE PERFORMED:</b>    | 09/24/2026                            | <b>DATE SUBMITTED:</b> | 09/24/2026            |
| <b>NAME:</b>              | Galan, Irick Marvin A.                | <b>INSTRUCTOR:</b>     | Engr. Jimlord Quejado |

|                       |                              |
|-----------------------|------------------------------|
| <b>MIDTERM SKILLS</b> | <b>Installing Prometheus</b> |
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### 1. PROCEDURE OUTPUT

Sample output (students are allowed to add more steps as needed.)

Task 1: Prepare the Server

Create first the monitoring roles directory using **mkdir -p roles/monitoring/tasks** under your desired directory

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS
qimagalan@workstation: ~ x qimagalan@workstation: ~/CPE212_Galan/MI... x
qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS$ ls -R
.:
install_prometheus.yml  roles

./roles:
monitoring

./roles/monitoring:
tasks

./roles/monitoring/tasks:
qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS$
```

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS
qimagalan@workstation: ~/CPE212_Galan/HO... x qimagalan@workstation: ~/CPE212_Galan/MI... x
GNU nano 7.2 roles/monitoring/tasks/main.yml
---
- name: Prepare the Servers (Ubuntu)
  ansible.builtin.apt:
    update_cache: true
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: Prepare the Servers (CentOS)
  ansible.builtin.dnf:
    update_cache: true
    state: latest
  when: ansible_distribution == "CentOS"
```

Create the main\_installer.yml to run the role

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS
qimagalan@workstation: ~/CPE212_Galan/HO... x qimagalan@workstation: ~/CPE212_Galan/MI... x
GNU nano 7.2 main_installer.yml
---
- hosts:
  - web_servers
  - file_servers
  become: true
  roles: monitoring
```

Results;

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS
qimagalan@workstation: ~... x qimagalan@workstation: ~... x qimagalan@workstation: ~... x
lller.yml -i inventory.yml -K
BECOME password:

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Prepare the Servers (Ubuntu)] *****
skipping: [192.168.56.129]
changed: [192.168.56.127]

TASK [monitoring : Prepare the Servers (CentOS)] *****
skipping: [192.168.56.127]
ok: [192.168.56.129]

PLAY RECAP *****
192.168.56.127      : ok=2    changed=1    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0
192.168.56.129      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0

qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS$
```

Task 2: Create the Prometheus User

```
- name: Create the Prometheus User
  ansible.builtin.user:
    name: prometheus
    home: /home/prometheus
    create_home: true
    shell: /sbin/nologin
    system: true

[ Wrote 22 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Run and Check Results on Servers:

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS
TASK [Gathering Facts] *****
ok: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Prepare the Servers (Ubuntu)] *****
skipping: [192.168.56.129]
changed: [192.168.56.127]

TASK [monitoring : Prepare the Servers (CentOS)] *****
skipping: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Create the Prometheus User] *****
changed: [192.168.56.127]
changed: [192.168.56.129]

PLAY RECAP *****
192.168.56.127      : ok=3    changed=2    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0
192.168.56.129      : ok=3    changed=1    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0

qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS$
```

Check by ssh through servers and run **cat /etc/passwd | grep prometheus**

```
qimagalan@GalanCentOS:~
qimagalan@server1:~$ cat /etc/passwd | grep prometheus
prometheus:x:995:983:~/home/prometheus:/sbin/nologin
qimagalan@server1:~$
logout
Connection to server1 closed.
qimagalan@workstation:~/CPE212_Galan/H0A8$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Sep 23 20:27:08 2026 from 192.168.56.126
[qimagalan@GalanCentOS ~]$ cat /etc/passwd | grep prometheus
prometheus:x:983:983:~/home/prometheus:/sbin/nologin
[qimagalan@GalanCentOS ~]$
```

### Task 3: Create Prometheus Directories

```
qimagalan@workstation: ~... x qimagalan@workstation: ~... x qimagalan@workstation: ~... x
GNU nano 7.2 roles/monitoring/tasks/main.yml

- name: Create the Prometheus User
  ansible.builtin.user:
    name: prometheus
    home: /home/prometheus
    create_home: true
    shell: /sbin/nologin
    system: true
    state: present

- name: Create the necessary directories for prometheus
  ansible.builtin.file:
    path:
      - "/etc/prometheus"
      - "/var/lib/prometheus"
    owner: prometheus
    group: prometheus
    mode: "0644"
    state: directory

[ Wrote 32 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
qimagalan@workstation: ~... x qimagalan@workstation: ~... x qimagalan@workstation: ~... x
GNU nano 7.2 monitoring/tasks/main.yml
name: prometheus
home: /home/prometheus
create_home: true
shell: /sbin/nologin
system: true
state: present

- name: Create the necessary directories for prometheus
  ansible.builtin.file:
    path: "[[ item ]]"
    owner: prometheus
    group: prometheus
    mode: "0644"
    state: directory
    loop:
      - "/etc/prometheus"
      - "/var/lib/prometheus"

[ Wrote 33 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

*Run and Check Results:*

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS
qimagalan@workstation: ~... x qimagalan@workstation: ~... x qimagalan@server1: ~ x v
skipping: [192.168.56.129]
changed: [192.168.56.127]

TASK [monitoring : Prepare the Servers (CentOS)] *****
skipping: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Create the Prometheus User] *****
ok: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Create the necessary directories for prometheus] *****
changed: [192.168.56.127] => (item=/etc/prometheus)
changed: [192.168.56.129] => (item=/etc/prometheus)
changed: [192.168.56.127] => (item=/var/lib/prometheus)
changed: [192.168.56.129] => (item=/var/lib/prometheus)

PLAY RECAP *****
192.168.56.127 : ok=4 changed=2 unreachable=0 failed=0 s
kipped=1 rescued=0 ignored=0
192.168.56.129 : ok=4 changed=1 unreachable=0 failed=0 s
kipped=1 rescued=0 ignored=0

qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS$

13 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Sep 24 08:44:40 2026 from 192.168.56.126
qimagalan@server1:~$ ls -l /etc | grep prometheus
drw-r--r-- 2 prometheus prometheus 4096 Sep 24 08:44 prometheus

qimagalan@server1:~$ ls -l /var/lib | grep prometheus
drw-r--r-- 2 prometheus prometheus 4096 Sep 24 08:44 prometheus

qimagalan@server1:~$
```

```
qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS/roles$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Sep 23 20:44:42 2026 from 192.168.56.126
[qimagalan@GalanCentOS ~]$ ls -l /etc | grep prometheus
drw-r--r--. 2 prometheus prometheus 6 Sep 23 20:44 prometheus
[qimagalan@GalanCentOS ~]$ ls -l /var/lib | grep prometheus
drw-r--r--. 2 prometheus prometheus 6 Sep 23 20:44 prometheus
[qimagalan@GalanCentOS ~]$
```

#### Task 4: Download Prometheus

```
qimagalan@workstation: ~/CPE212_Galan/HOA8 x qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKI... x qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKI... x
GNU nano 7.2 monitoring/tasks/main.yml
system: true
state: present

- name: Create the necessary directories for prometheus
  ansible.builtin.file:
    path: "[[ item ]]"
    owner: prometheus
    group: prometheus
    mode: "0644"
    state: directory
  loop:
    - "/etc/prometheus"
    - "/var/lib/prometheus"

- name: Download the prometheus package from link
  become: false
  ansible.builtin.get_url:
    url: "https://github.com/prometheus/prometheus/releases/download/v3.15.0-rc.0/prometheus-3.15.0-rc.0.linux-amd64.tar.gz"
    dest: "[[ ansible_env.HOME ]]/Downloads"
```

```
qimagalan@workstation: ~/CPE212_Galan/HOA8 x qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKI... x qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKI... x
TASK [monitoring : Prepare the Servers (Centos)] *****
skipping: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Create the Prometheus User] *****
ok: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Create the necessary directories for prometheus] *****
ok: [192.168.56.127] => (item=/etc/prometheus)
ok: [192.168.56.129] => (item=/etc/prometheus)
ok: [192.168.56.127] => (item=/var/lib/prometheus)
ok: [192.168.56.129] => (item=/var/lib/prometheus)

TASK [monitoring : Download the prometheus package from link] *****
changed: [192.168.56.127]
changed: [192.168.56.129]

PLAY RECAP *****
192.168.56.127 : ok=5 changed=2 unreachable=0 failed=0 skipped=1 rescued=0 ignored=0
192.168.56.129 : ok=5 changed=1 unreachable=0 failed=0 skipped=1 rescued=0 ignored=0

qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS$
```

```
qimagalan@workstation: ~/CPE212_Galan/HOA8 x qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKI... x

qimagalan@server1:~$ ls ~/Downloads
prometheus-3.15.0-rc.0.linux-amd64.tar.gz
qimagalan@server1:~$
logout
Connection to server1 closed.
qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS/roles$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Sep 23 20:55:11 2026 from 192.168.56.126
[qimagalan@GalanCentOS ~]$ ls ~/Downloads
prometheus-3.15.0-rc.0.linux-amd64.tar.gz
[qimagalan@GalanCentOS ~]$
```

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS/roles
qimagalan@server1: ~ x qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKI... x qimagalan@workstation: ~/CPE212_Ga

GNU nano 7.2 monitoring/tasks/main.yml
- "/etc/prometheus"
- "/var/lib/prometheus"

- name: Download the prometheus package from link
  become: false
  ansible.builtin.get_url:
    url: "https://github.com/prometheus/prometheus/releases/download/v3.13.3/prometheus-3.13.3.linux-amd64.tar.gz"
    dest: "/home/qimagalan/Downloads"

- name: Create temp directory for prometheus installation
  ansible.builtin.file:
    path: "/home/qimagalan/prometheus"
    state: directory

- name: Extract the tar files
  ansible.builtin.unarchive:
    src: "/home/qimagalan/Downloads/prometheus-3.13.3.linux-amd64.tar.gz"
    dest: "/home/qimagalan/prometheus"
    remote_src: true

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo     M-6 Copy
```



```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS

TASK [monitoring : Create the necessary directories for prometheus] *****
ok: [192.168.56.127] => (item=/etc/prometheus)
ok: [192.168.56.129] => (item=/etc/prometheus)
ok: [192.168.56.127] => (item=/var/lib/prometheus)
ok: [192.168.56.129] => (item=/var/lib/prometheus)

TASK [monitoring : Download the prometheus package from link] *****
changed: [192.168.56.129]
changed: [192.168.56.127]

TASK [monitoring : Create temp directory for prometheus installation] *****
changed: [192.168.56.129]
ok: [192.168.56.129]

TASK [monitoring : Extract the tar files] *****
changed: [192.168.56.129]
changed: [192.168.56.127]

PLAY RECAP *****
192.168.56.127      : ok=7    changed=4    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.129      : ok=7    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS$

qimagalan@server1:~$ ls ~/prometheus
prometheus-3.13.3.linux-amd64
qimagalan@server1:~$
logout
Connection to server1 closed.
qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS/roles$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Sep 23 21:18:27 2026 from 192.168.56.126
[qimagalan@GalanCentOS ~]$ ls ~/prometheus
prometheus-3.13.3.linux-amd64  prometheus-3.15.0-rc.0.linux-amd64
[qimagalan@GalanCentOS ~]$
```

Task 5: Install the Prometheus Binaries

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SK
qimagalan@server1: ~ × qimagalan@workstati... × qimagalan@workstati... × qimagalan@workstati...
GNU nano 7.2 roles/monitoring/tasks/main.yml

- name: Copy the prometheus bin
  become: true
  ansible.builtin.copy:
    src: "{{ ansible_env.HOME }}/prometheus/prometheus-3.13.3.linux-amd64/prometheus"
    dest: "/usr/local/bin/prometheus"
    mode: "0644"
    owner: prometheus
    group: prometheus
    remote_src: true

- name: Copy the promtool bin
  become: true
  ansible.builtin.copy:
    src: "{{ ansible_env.HOME }}/prometheus/prometheus-3.13.3.linux-amd64/promtool"
    dest: "/usr/local/bin/promtool"
    mode: "0644"
    owner: prometheus
    group: prometheus
    remote_src: true

- name: Copy the configuration file to /etc
  become: true
  ansible.builtin.copy:
    src: "{{ ansible_env.HOME }}/prometheus/prometheus-3.13.3.linux-amd64/prometheus.yml"
    dest: "/etc/prometheus/prometheus.yml"
    mode: "0644"
    owner: prometheus
    group: prometheus
    remote_src: true
```

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_5
qimagalan@server1: ~ × qimagalan@workstati... × qimagalan@workstati... × qimagalan@workstati...

TASK [monitoring : Create temp directory for prometheus installation] *****
ok: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Extract the tar files] *****
ok: [192.168.56.127]
ok: [192.168.56.129]

TASK [monitoring : Copy the prometheus bin] *****
changed: [192.168.56.129]
changed: [192.168.56.127]

TASK [monitoring : Copy the promtool bin] *****
changed: [192.168.56.127]
changed: [192.168.56.129]

TASK [monitoring : Copy the configuration file to /etc] *****
changed: [192.168.56.127]
changed: [192.168.56.129]
```

#### Task 6: Install the Prometheus Configuration Files

```
- name: Procure the consoles and console_libraries with apt
  ansible.builtin.apt:
    name: prometheus
    state: present
    when: ansible_distribution == "Ubuntu"

- name: Procure the consoles and console_libraries with dnf
  ansible.builtin.dnf:
    name: prometheus
    state: present
    when: ansible_distribution == "CentOS"
```

<sup>^G</sup> Help    <sup>^O</sup> Write Out    <sup>^W</sup> Where Is    <sup>^K</sup> Cut    <sup>^T</sup> Execute    <sup>^C</sup> Location  
<sup>^X</sup> Exit    <sup>^R</sup> Read File    <sup>^\_</sup> Replace    <sup>^U</sup> Paste    <sup>^J</sup> Justify    <sup>^/</sup> Go To Lin

#### Task 7: Validate the Configuration

*prometheus -config.file=/etc/prometheus/prometheus.yml*



```
qinagalan@server1:~$ prometheus --config.file=/etc/prometheus/prometheus.yml
time=2026-09-24T10:21:26.098+08:00 level=INFO source=main.go:1730 msg="updated GOGC" old=100 new=75
time=2026-09-24T10:21:26.099+08:00 level=INFO source=main.go:789 msg="Leaving GOMAXPROCS=4: CPU quota undefined" component=automaxpro
time=2026-09-24T10:21:26.101+08:00 level=INFO source=memlimit.go:198 msg="GOMEMLIMIT is updated" component=automemlimit package=githu
imMachineGun/automemlimit/memlimit GOMEMLIMIT=1806944256 previous=9223372036854775807
time=2026-09-24T10:21:26.102+08:00 level=INFO source=main.go:896 msg="Starting Prometheus Server" mode=server version="(version=3.13.
ch=HEAD, revision=b273ae3adeb64ad630d65ef7f16440df95658410)"
time=2026-09-24T10:21:26.102+08:00 level=INFO source=main.go:901 msg="operational information" build_context="(go=go1.26.8, platform=
md64, user=root@a29bd0364bed, date=20260907-11:42:33, tags=netgo,builtinassets)" host_details="(Linux 7.0.0-30-generic #30-24.04.1-Ub
P PREEMPT_DYNAMIC Fri Aug 7 13:27:52 UTC 2 x86_64 server1 (none))" fd_limits="(soft=1048575, hard=1048576)" vm_limits="(soft=unlimit
d=unlimited)"
time=2026-09-24T10:21:26.745+08:00 level=INFO source=web.go:727 msg="Start listening for connections" component=web address=0.0.0.0:9
time=2026-09-24T10:21:26.747+08:00 level=INFO source=main.go:1462 msg="Starting TSDB ..."
time=2026-09-24T10:21:26.755+08:00 level=INFO source=tsdb_config.go:354 msg="Listening on" component=web address=[::]:9090
time=2026-09-24T10:21:26.756+08:00 level=INFO source=tsdb_config.go:357 msg="TLS is disabled." component=web http2=false address=[::]:
time=2026-09-24T10:21:26.765+08:00 level=INFO source=head.go:709 msg="Replaying on-disk memory mappable chunks if any" component=tsdb
time=2026-09-24T10:21:26.766+08:00 level=INFO source=head.go:795 msg="On-disk memory mappable chunks replay completed" component=tsdb
on=3.88µs
time=2026-09-24T10:21:26.766+08:00 level=INFO source=head.go:803 msg="Replaying WAL, this may take a while" component=tsdb
time=2026-09-24T10:21:26.768+08:00 level=INFO source=head.go:898 msg="WAL segment loaded" component=tsdb segment=0 maxSegment=0 durat
18982ms
time=2026-09-24T10:21:26.768+08:00 level=INFO source=head.go:935 msg="WAL replay completed" component=tsdb checkpoint_replay_duration

[qinagalan@GalanCentOS ~]$ prometheus --config.file=/etc/prometheus/prometheus.yml
time=2026-09-23T22:22:14.917-04:00 level=INFO source=main.go:1730 msg="updated GOGC" old=100 new=75
time=2026-09-23T22:22:14.918-04:00 level=INFO source=main.go:789 msg="Leaving GOMAXPROCS=4: CPU quota undefined" component=automaxpro
time=2026-09-23T22:22:14.919-04:00 level=INFO source=memlimit.go:198 msg="GOMEMLIMIT is updated" component=automemlimit package=githu
imMachineGun/automemlimit/memlimit GOMEMLIMIT=1565363404 previous=9223372036854775807
time=2026-09-23T22:22:14.919-04:00 level=INFO source=main.go:896 msg="Starting Prometheus Server" mode=server version="(version=3.13.
ch=HEAD, revision=b273ae3adeb64ad630d65ef7f16440df95658410)"
time=2026-09-23T22:22:14.919-04:00 level=INFO source=main.go:901 msg="operational information" build_context="(go=go1.26.8, platform=
md64, user=root@a29bd0364bed, date=20260907-11:42:33, tags=netgo,builtinassets)" host_details="(Linux 5.14.0-741.el9.x86_64 #1 SMP PR
YNAMIC Fri Aug 21 18:33:15 UTC 2026 x86_64 GalanCentOS (none))" fd_limits="(soft=524287, hard=524288)" vm_limits="(soft=unlimited, ha
mited)"
time=2026-09-23T22:22:15.339-04:00 level=INFO source=web.go:727 msg="Start listening for connections" component=web address=0.0.0.0:9
time=2026-09-23T22:22:15.342-04:00 level=INFO source=main.go:1462 msg="Starting TSDB ..."
time=2026-09-23T22:22:15.345-04:00 level=INFO source=tsdb_config.go:354 msg="Listening on" component=web address=[::]:9090
time=2026-09-23T22:22:15.345-04:00 level=INFO source=tsdb_config.go:357 msg="TLS is disabled." component=web http2=false address=[::]:
time=2026-09-23T22:22:15.355-04:00 level=INFO source=head.go:709 msg="Replaying on-disk memory mappable chunks if any" component=tsdb
time=2026-09-23T22:22:15.356-04:00 level=INFO source=head.go:795 msg="On-disk memory mappable chunks replay completed" component=tsdb
on=3.767µs
time=2026-09-23T22:22:15.356-04:00 level=INFO source=head.go:803 msg="Replaying WAL, this may take a while" component=tsdb
time=2026-09-23T22:22:15.357-04:00 level=INFO source=head.go:898 msg="WAL segment loaded" component=tsdb segment=0 maxSegment=0 durat
.053µs
time=2026-09-23T22:22:15.357-04:00 level=INFO source=head.go:935 msg="WAL replay completed" component=tsdb checkpoint_replay_duration
```

## Task 8: Create the systemd Service

```
- name: Procure the consoles and console_libraries with apt
  become: true
  ansible.builtin.apt:
    name: prometheus
    state: present
  when: ansible_distribution == "Ubuntu"

- name: Procure the consoles and console_libraries with dnf
  become: true
  ansible.builtin.dnf:
    name: prometheus
    state: present
  when: ansible_distribution == "CentOS"
```

```
qimagalan@work... x qimagalan@work... x qimagalan@work... x qima
[qimagalan@GalanCentOS ~]$ ls /etc/systemd/system | grep prom
prometheus.service
[qimagalan@GalanCentOS ~]$
```

Check the consoles and console\_libraries directory

```
TASK [monitoring : Procure the consoles and console_libraries with dnf] *****
changed: [192.168.56.129]

TASK [monitoring : Create the daemon file] *****
changed: [192.168.56.129]

bash: console_libraries: command not found
[qimagalan@GalanCentOS ~]$ ls /etc/prometheus | grep -E "consoles|console_libraries"
console_libraries
consoles
[qimagalan@GalanCentOS ~]$
```

Task 9: Start and Enable Prometheus

```
- name: Restart the the systemd daemon
  become: true
  ansible.builtin.service:
    daemon_reload: true

- name: Start the prometheus service
  become: true
  ansible.builtin.service:
    name: prometheus
    started: true
    enabled: true
```

```
qimagalan@workstation: ~/CPE212_Galan/MIDTERM_SKILLS
qimagalan@serve... x qimagalan@work... x qimagalan@work... x qimagalan@work... x qimagalan@work... x qimagalan@work... x qimagalan@Gala...
skipping: [192.168.56.127]
ok: [192.168.56.127]

TASK [monitoring : Create the daemon file] *****
changed: [192.168.56.127]
ok: [192.168.56.127]

TASK [monitoring : Restart the the systemd daemon] *****
ok: [192.168.56.127]
ok: [192.168.56.127]

TASK [monitoring : Start the prometheus service] *****
fatal: [192.168.56.127]: FAILED! => {"changed": false, "msg": "Unsupported parameters for (ansible.legacy.systemd) module: started. S
d parameters include: daemon_reexec, daemon_reload, enabled, force, masked, name, no_block, scope, state (daemon-reexec, daemon-reloa
ice, unit)."}
fatal: [192.168.56.127]: FAILED! => {"changed": false, "msg": "Unsupported parameters for (ansible.legacy.systemd) module: started. S
d parameters include: daemon_reexec, daemon_reload, enabled, force, masked, name, no_block, scope, state (daemon-reexec, daemon-reloa
ice, unit)."}

PLAY RECAP *****
192.168.56.127      : ok=11  changed=1  unreachable=0    failed=1    skipped=2    rescued=0    ignored=0
192.168.56.129      : ok=13  changed=0  unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

qimagalan@workstation:~/CPE212_Galan/MIDTERM_SKILLS$
```

Task 10: Verify Prometheus  
netstat -tln | grep 9090

prometheus --version

```
qimagalan@server1:~$ prometheus --version
prometheus, version 3.13.3 (branch: HEAD, revision: b273ae3adeb64ad630d65ef7f16440df95658410)
  build user:      root@a29bd0364bed
  build date:      20260907-11:42:33
  go version:      go1.26.8
  platform:        linux/amd64
  tags:            netgo,builtinassets
qimagalan@server1:~$

qimagalan@GalanCentOS:~$ prometheus --version
prometheus, version 3.13.3 (branch: HEAD, revision: b273ae3adeb64ad630d65ef7f16440df95658410)
  build user:      root@a29bd0364bed
  build date:      20260907-11:42:33
  go version:      go1.26.8
  platform:        linux/amd64
  tags:            netgo,builtinassets
qimagalan@GalanCentOS:~$
```

Task 11: Access Prometheus Through a Browser

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:



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---

**Tools Used:**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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